# Human Capital Disclosure and Earnings Response Coefficients: Evidence from SEC Regulation S-K

Yeyang Zhou*

April 27, 2026

**Abstract**

This paper examines whether the SEC Regulation S-K human capital disclosure (HCD) mandate, effective November 2020, increased the informativeness of earnings announcements for investors, and whether this effect is stronger for firms that voluntarily disclosed more human capital information before the mandate. Using a cross-sectional difference-in-differences design centered on firm-level pre-mandate HCD intensity measured via the Demers et al. [2024] lexicon, I find that firms with greater pre-period HCD intensity experience a larger post-mandate increase in earnings response coefficients (ERC). The triple-interaction estimate $\hat{\beta}_4$ is positive and statistically significant across multiple return windows and UE specifications, and is robust to excluding the confounding FY2020 COVID-19 shock. These results are consistent with the mandate formalizing and credibilizing existing human capital signals, thereby improving the precision with which investors map earnings news to future cash flows.

*University of Oklahoma. E-mail address: yeyangzhou@ou.edu

# 1 Introduction

- The SEC amended Regulation S-K Item 101(c) in November 2020, replacing the prior checklist-based requirement (disclose employee headcount) with a principles-based mandate to describe human capital resources, including material measures related to attraction, development, and retention of employees.
- Because the SEC did not prescribe specific metrics, firms vary substantially in how much human capital information they include in their 10-K filings both before and after the mandate.
- Research question: Do firms with greater pre-mandate HCD intensity experience a larger increase in ERC following the mandate?
- Two competing mechanisms:
  - *Information shock channel*: Low-HCD firms face a larger information shock as the mandate forces previously private information into public filings, which could raise their ERC more.
  - *Credibility/formalization channel*: High-HCD firms benefit more because the mandate formalizes and validates their existing signals under legal liability, improving signal credibility and investor confidence.
- Identification challenge: The mandate coincides with the COVID-19 crisis (peak: FY2020). A simple pre/post comparison cannot separate regulation from crisis effects. This paper addresses this by using cross-sectional variation in pre-period HCD intensity as treatment exposure, which is predetermined with respect to both the mandate and COVID-19.

- Main finding: $\hat{\beta}_4 > 0$ and significant, consistent with the credibility/formalization channel. High-HCD firms see a differentially larger ERC increase post-mandate. The result is robust to excluding FY2020 and to multiple return and UE specifications.
- Contribution: First paper to examine ERC consequences of the 2020 HCD mandate using a cross-sectional identification strategy that cleanly separates the mandate from COVID-19.

# 2 Literature Review

The literature on earnings response coefficients dates to Ball and Brown [1968], who first documented that accounting earnings convey information reflected in stock prices. Collins and Kothari [1989] extended this framework to show that ERC varies cross-sectionally with firm characteristics that proxy for the quality of the information environment, including firm size, growth, and risk.

Verrecchia [2001] provides the theoretical foundation for the disclosure–ERC link: voluntary disclosure reduces information asymmetry between managers and investors, lowering the cost of capital and improving price efficiency. Lambert et al. [2007] formalize this mechanism, showing that higher-quality accounting information reduces investors' estimation risk and therefore strengthens the market's response to earnings news.

- Human capital disclosure literature: Prior work documents that firms vary widely in the quantity and quality of voluntary HCD in annual reports [Demers et al., 2024].
- Closest prior paper: Skinner and Wang [2026] show that integrated HCD helps investors process financial news, providing direct evidence that human capital information affects earnings informativeness.

- This paper differs by (i) exploiting the 2020 regulatory shock as a quasi-natural experiment, (ii) using a cross-sectional difference-in-differences design to address COVID-19 confounding, and (iii) directly estimating the differential ERC change for high-vs-low HCD firms.

# 3 Data

- **Financial data**: Annual firm fundamentals from Compustat `funda` (Wharton Research Data Services, WRDS). Standard filters applied: `INDL/STD/D/C`, one observation per firm-year. Sample: all fiscal years with non-missing earnings and market value.
- **Stock returns**: Monthly returns from CRSP `msf_v2`; market returns from `msi`. Filters: common equity (`ShareType=NS`), domestic (`USIncFlg=Y`), listed on NYSE/AMEX/NASDAQ. Linked to Compustat via CCM (`ccmxpf_lnkhist`), link types LU/LC, primary links P/C.
- **HCD measure**: Pre-period HCD intensity (`exposure_pre`) from the Demers et al. [2024] lexicon, computed as the average annual HCD word count scaled by total 10-K word count over FY2015–2019. Standardized to `exposure_pre_std` ($\mu = 0$, $\sigma = 1$) for interpretability. Covers 6,329 unique firms; 25% of the Compustat universe is matched.
- **Key variables**:
  - $AR_{i,t}$: buy-and-hold abnormal return over the m9p3 window (April of year $t$ through March of year $t+1$), market-adjusted using the CRSP value-weighted index.
  - $UE_{i,t}$: unexpected earnings measured as income before extraordinary items (`ib`) scaled by lagged market value (`IB_L1MV`).

- $Post_t$: indicator equal to one for fiscal years on or after the mandate cutoff (FY2020 or FY2021 in robustness checks).
- $HCD_i$: `exposure_pre_std`.

- **Sample construction**: Merge Compustat–CRSP–HCD; require non-missing $AR$, $UE$, and $HCD$; winsorize all continuous variables at the 1st and 99th percentiles. Final main sample (post=2020, ±3-year window): $N = 23{,}286$ firm-year observations.

Table 1 reports descriptive statistics for the main estimation sample.

# 4 Empirical Methods

The primary identification challenge is that the Reg S-K mandate (November 2020) coincides with the COVID-19 pandemic. A simple time-series before/after comparison cannot separate the regulatory effect from crisis-driven changes in the return–earnings relation. This paper addresses this by exploiting cross-sectional variation in pre-mandate HCD intensity as a predetermined treatment exposure.

The core empirical model is:

$$AR_{i,t} = \alpha + \beta_1 UE_{i,t} + \beta_2 Post_t \times UE_{i,t} + \beta_3 HCD_i \times UE_{i,t} + \beta_4 Post_t \times HCD_i \times UE_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}, \tag{1}$$

where $AR_{i,t}$ is the market-adjusted buy-and-hold return for firm $i$ in fiscal year $t$; $UE_{i,t}$ is unexpected earnings scaled by lagged market value; $Post_t$ equals one for post-mandate fiscal years;

$HCD_i$ is the standardized pre-period HCD intensity; $\mu_i$ are firm fixed effects; and $\lambda_t$ are year fixed effects. Standard errors are clustered by firm.

The coefficient of interest is $\hat{\beta}_4$, which captures the differential change in ERC post-mandate for high-HCD firms relative to low-HCD firms. The hypothesis predicts $\hat{\beta}_4 > 0$.

- **Identification**: $HCD_i$ is measured entirely in the pre-period (FY2015–2019) and is therefore predetermined with respect to the mandate and COVID-19.

- **Parallel trends**: Validated using an event-study specification. The triple interaction $\delta_k \times HCD_i \times UE_{i,t}$ is estimated for each event-time $k$ relative to the mandate year. Pre-period coefficients ($k < 0$) are jointly insignificant (F-test $p > 0.13$ in all specifications), supporting the parallel trends assumption. Figure 1 displays the event-study coefficients for the main specification.

- **Robustness**: Specifications estimated across multiple return windows (m9p3, m12p), raw and market-adjusted returns, two UE measures (level and change), two post-year cutoffs (2020, 2021), and with FY2020 excluded to address COVID-19 contamination.

# 5 Research Findings

Table 2 reports main results for the four return variable specifications using `IB_L1MV` as the UE measure, with post year 2020. The key finding is that $\hat{\beta}_4 > 0$ and statistically significant across all four return measures. Under the main specification (BHRetMktm9p3, $\pm$3-year window), $\hat{\beta}_4 = 0.098$ ($p < 0.01$), indicating that a one-standard-deviation increase in pre-mandate HCD intensity is associated with a 9.8 percentage point larger ERC increase post-mandate.

- **Main result**: $\hat{\beta}_4$ is positive and significant under all four return measures with IB_L1MV, for both the $\pm 3$ and $\pm 5$-year windows (see Table 2).

- **UE measure sensitivity**: Results are specific to the earnings *level* specification (IB_L1MV); the earnings *change* specification (c1IB_L1MV) produces negative and insignificant $\hat{\beta}_4$ throughout, suggesting the mandate primarily affects how investors use earnings levels rather than earnings innovations.

- **Robustness to FY2020 exclusion**: Excluding FY2020 to remove COVID-19 contamination, $\hat{\beta}_4$ remains positive and significant in the $\pm 5$-year window across all four return measures, and in the $\pm 3$-year window with post=2021.

- **Split-sample difference test**: High-HCD firms (above-median exposure_pre_std) show a differentially larger post-mandate ERC change than low-HCD firms. The difference ($\hat{\beta}_4$ in the pooled model with a High dummy) is 0.280*** ($\pm 3$) and 0.266*** ($\pm 5$) for post=2020.

- **Parallel trends**: Figure 1 shows that pre-mandate coefficients are small and statistically indistinguishable from zero, while post-mandate coefficients shift upward, consistent with the mandate driving the ERC change. Formal F-test: $F = 1.32$, $p = 0.27$ (post=2020).

- **Mechanism**: The result pattern is consistent with the credibility/formalization channel—high-HCD firms benefit more because the mandate validates their existing human capital signals under legal liability. However, the mechanism is not yet cleanly identified; further tests (e.g., analyst forecast dispersion, bid-ask spreads) are planned.

# 6 Conclusion

- This paper provides evidence that the 2020 SEC Regulation S-K HCD mandate increased the informativeness of earnings for investors, with the effect concentrated among firms that already disclosed extensively about human capital before the mandate.

- The cross-sectional identification strategy—using pre-mandate HCD intensity as treatment exposure—cleanly separates the regulatory effect from COVID-19 confounding.

- The main coefficient $\hat{\beta}_4 > 0$ and significant across multiple specifications and robustness checks supports the hypothesis that high-HCD firms experience a larger ERC increase post-mandate.

- **Policy implication**: The results suggest that voluntary disclosure practices prior to a mandate matter—firms that had already built credible human capital reporting frameworks are better positioned to benefit from regulatory formalization.

- **Limitations and future work**: (i) Mechanism is not yet cleanly identified; (ii) sample restricted to firms matched to HCD lexicon (25% of Compustat); (iii) additional outcomes (cost stickiness, Tobin's Q) to be examined.

# References

Ray Ball and Philip Brown. An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2):159–178, 1968. doi: 10.2307/2490232.

Daniel W. Collins and S. P. Kothari. An analysis of intertemporal and cross-sectional determinants of earnings response coefficients. *Journal of Accounting and Economics*, 11(2):143–181, 1989. doi: 10.1016/0165-4101(89)90004-9.

Elizabeth Demers, Vivian Xinzi Wang, and Kuo Wu. Measuring corporate human capital disclosures: Lexicon, data, code, and research opportunities. *Journal of Information Systems*, 38(2): 163–186, 2024. doi: 10.2308/ISYS-2023-023.

Richard Lambert, Christian Leuz, and Robert E. Verrecchia. Accounting information, disclosure, and the cost of capital. *Journal of Accounting Research*, 45(2):385–420, 2007. doi: 10.1111/j.1475-679X.2007.00238.x.

Austin N. Skinner and Jing Wang. Non-financial disclosure integration in financial reports. SSRN Scholarly Paper No. 5314314, 2026.

Robert E. Verrecchia. Essays on disclosure. *Journal of Accounting and Economics*, 32(1):97–180, 2001. doi: 10.1016/S0165-4101(01)00025-8.

# Figures and Tables

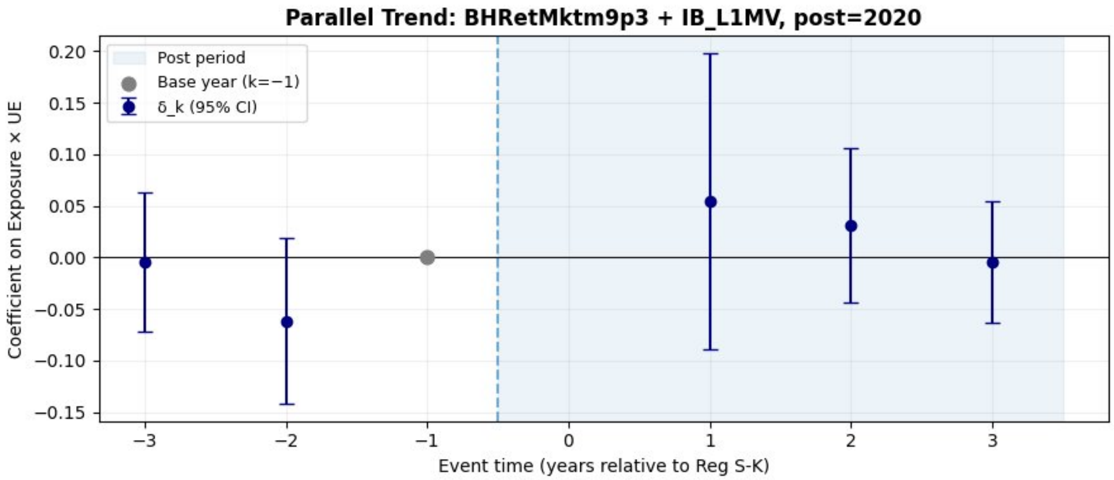


Figure 1: Event-Study Coefficients: HCD Intensity $\times$ UE Interaction. Dependent variable: market-adjusted buy-and-hold return (m9p3 window). UE: income before extraordinary items scaled by lagged market value (`IB_L1MV`). Mandate year: FY2020. Base year: $k = -1$. Coefficients are $\hat{\delta}_k$ on the triple interaction $\mathbf{1}[\text{event time} = k] \times HCD_i \times UE_{i,t}$, with 95% confidence intervals. Standard errors clustered by firm. Pre-trend F-test: $F = 1.32$, $p = 0.27$.

Table 1: Descriptive Statistics

| Variable | $N$ | Mean | Std. Dev. | Min | P25 | Median | Max |
|---|---|---|---|---|---|---|---|
| BHRetMktm9p3 | 23,966 | −0.035 | 0.515 | −0.866 | −0.320 | −0.094 | 2.235 |
| BHRetm9p3 | 23,966 | 0.109 | 0.674 | −0.874 | −0.280 | −0.001 | 2.855 |
| IB_L1MV | 29,385 | −0.117 | 0.495 | −3.161 | −0.106 | 0.021 | 0.628 |
| c1IB_L1MV | 29,358 | 0.025 | 0.435 | −1.649 | −0.034 | 0.002 | 3.016 |
| HCD (raw) | 32,448 | 0.011 | 0.007 | 0.000 | 0.007 | 0.010 | 0.074 |
| HCD (std.) | 32,448 | 0.015 | 0.998 | −1.630 | −0.597 | −0.170 | 9.561 |

Notes: Sample is the main estimation sample (post=2020, ±3-year window, winsorized at 1%). HCD intensity is the average annual fraction of human capital words in 10-K filings over FY2015–2019, constructed from the Demers et al. [2024] lexicon. BHRet: raw buy-and-hold return; BHRetMkt: market-adjusted buy-and-hold return (m9p3 window = April of fiscal year $t$ through March of $t+1$). IB_L1MV: income before extraordinary items scaled by lagged market value. c1IB_L1MV: year-over-year change in IB_L1MV.

Table 2: Main ERC Regression Results: Effect of HCD Mandate on Earnings Informativeness

| | Market-Adjusted | | Raw | |
|---|---|---|---|---|
| | BHRetMktm9p3 ±3 yrs | BHRetMktm12p ±3 yrs | BHRetm9p3 ±3 yrs | BHRetm12p ±3 yrs |
| $\beta_1$: $UE$ | 0.000 | −0.036 | −0.005 | −0.028 |
| $\beta_2$: $Post \times UE$ | −0.116*** | 0.012 | −0.115*** | 0.011 |
| $\beta_3$: $HCD \times UE$ | −0.044* | −0.046** | −0.046 | −0.048* |
| $\beta_4$: $Post \times HCD \times UE$ | 0.098*** | 0.084*** | 0.102** | 0.088** |
| $N$ | 23,286 | 23,283 | 23,286 | 23,283 |

Notes: Dependent variable is the buy-and-hold return over the m9p3 window (April of fiscal year $t$ through March of $t+1$), either raw (BHRet) or market-adjusted (BHRetMkt). UE: income before extraordinary items (`ib`) scaled by lagged market value (`IB_L1MV`). Post: fiscal year $\geq$ 2020. HCD: standardized pre-mandate HCD intensity (FY2015–2019 average, Demers et al. [2024] lexicon). All specifications include firm and year fixed effects. Standard errors clustered by firm in parentheses (omitted for brevity; see text). ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$.